

# Telco Customer Churn Analysis

Business Analytics Case Study Report

Future Interns Program | Data Analytics Track | 2025

**7,043**  
Total Customers

**1,869**  
Churned Customers

**26.5%**  
Churn Rate

**\$64.76**  
Avg Monthly Charge

## 1. Introduction

Customer churn — the rate at which customers stop doing business with a company — is one of the most critical metrics in the telecommunications industry. Acquiring a new customer is significantly more expensive than retaining an existing one, making churn prediction and prevention a top business priority. This report presents an end-to-end analysis of customer churn data from a telecommunications company, identifying the key drivers of churn and translating those findings into actionable business recommendations.

## 2. Business Problem Statement

The telecommunications company is experiencing a churn rate of 26.5%, meaning more than one in four customers leaves within any given period. This level of attrition poses a significant threat to revenue stability and long-term growth. The business needs to understand:

- Which customer segments are most likely to churn?
- What contract, service, and payment characteristics are associated with higher churn?
- What targeted interventions can reduce churn and improve customer lifetime value?

## 3. Dataset Description

The dataset used in this analysis is the IBM Watson Telco Customer Churn dataset, a widely used benchmark dataset in customer analytics. It contains 7,043 records and 21 columns covering four broad categories of customer information:

| Category        | Columns                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Demographics    | Gender, SeniorCitizen, Partner, Dependents                                                                                              |
| Services        | PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies |
| Account Info    | Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges, Tenure                                                         |
| Target Variable | Churn (Yes / No)                                                                                                                        |

## 4. Data Analysis Process

---

### Data Loading & Inspection

The raw CSV was loaded into a pandas DataFrame. Initial inspection confirmed 7,043 rows and 21 columns with no duplicate records. The TotalCharges column was identified as an object type despite containing numeric data.

### Exploratory Data Analysis (EDA)

Summary statistics, value counts, and missing value checks were performed across all columns before any cleaning took place, ensuring cleaning decisions were grounded in observed data rather than assumptions.

### Data Cleaning

TotalCharges was converted to numeric using `pd.to_numeric()` with `errors='coerce'`. The 11 resulting NaN values (belonging to new customers with tenure = 0) were filled with 0. A numeric ChurnFlag column (Yes=1, No=0) was created for aggregation. SeniorCitizen was re-labelled from 0/1 to 'Non-Senior'/'Senior' for readability. Tenure was grouped into four ordered bands: 0-12 mo, 13-24 mo, 25-48 mo, and 49-72 mo.

### Visualisation

Ten charts were produced in Python using Matplotlib and Seaborn, covering churn distribution, churn rate by contract, internet service, payment method, tenure bucket, senior citizen status, paperless billing, and add-on service subscriptions.

### Dashboard

An interactive Power BI dashboard was built from the cleaned dataset, featuring KPI cards and bar charts with DAX-calculated churn rate measures.

## 5. Key Findings

| #  | Finding                                | Churn Rate |
|----|----------------------------------------|------------|
| 1  | Month-to-month contract customers      | ~43%       |
| 2  | Fiber optic internet service customers | ~42%       |
| 3  | Electronic check payment users         | ~45%       |
| 4  | New customers (0–12 months tenure)     | ~48%       |
| 5  | Senior citizens                        | ~41%       |
| 6  | Customers without Online Security      | ~42%       |
| 7  | Customers without Tech Support         | ~42%       |
| 8  | Paperless billing customers            | ~33%       |
| 9  | Two-year contract customers            | ~3%        |
| 10 | Customers with 49–72 months tenure     | ~6%        |

## 6. Business Insights

### Contract Type is the Strongest Churn Predictor

The difference in churn rate between month-to-month (~43%) and two-year contracts (~3%) is the single largest gap in the entire dataset. Customers on flexible contracts face no switching cost, making them highly vulnerable to competitor offers.

### The First Year is the Most Critical Period

Customers in their first 12 months churn at nearly 48% — almost double the overall churn rate. This points to an onboarding and early engagement problem. Customers who make it past 24 months become significantly more loyal.

### Fiber Optic Customers Expect More

Despite being a premium service, fiber optic customers churn at a higher rate than DSL users. This suggests a gap between expectations and delivered service quality — likely related to pricing, speed consistency, or customer support responsiveness.

### Payment Method Reflects Customer Commitment

Electronic check users — typically customers who have not set up automatic payments — churn at the highest rate. Auto-pay customers (credit card, bank transfer) are significantly more retained, indicating that payment friction is also a loyalty signal.

### Add-On Services Act as Retention Anchors

Customers subscribed to Online Security and Tech Support churn at notably lower rates. These services increase switching costs and perceived value, making them powerful retention tools that are currently underutilised.

## 7. Recommendations

### 01 Incentivise Long-Term Contracts

Target month-to-month customers with targeted upgrade offers — discounted rates, waived setup fees, or bonus data — to migrate them to one or two-year contracts. Since month-to-month customers churn at 43% compared to 3% for two-year customers, even a modest conversion rate would have a significant impact on overall churn. Consider offering a loyalty discount in the 10th or 11th month as a proactive retention trigger before customers consider switching.

### 02 Implement a 90-Day Onboarding Programme

New customers (0–12 months) are the highest-risk segment. A structured onboarding programme — including a welcome call, proactive check-ins at 30, 60, and 90 days, and guided setup for add-on services — would significantly improve early retention. Assign dedicated support agents to new customers during their first three months to resolve issues before they become reasons to leave.

### 03 Address Fiber Optic Service Quality

Investigate the root causes of high churn among fiber optic customers through customer satisfaction surveys and support ticket analysis. If pricing is the issue, introduce a loyalty pricing tier for long-term fiber customers. If service quality is the issue, prioritise infrastructure investment and response time improvements for this segment. Fiber optic customers represent high-value accounts — retaining them should be a strategic priority.

### 04 Promote Auto-Pay Enrolment

Electronic check users churn at the highest rate of any payment segment. Launch a campaign to migrate these customers to automatic payment methods by offering a small monthly discount (e.g. \$2–5 off) for customers who switch to credit card or bank transfer auto-pay. This reduces payment friction, lowers churn risk, and improves cash flow predictability for the business.

### 05 Bundle Add-On Services as a Retention Strategy

Customers without Online Security and Tech Support churn at roughly twice the rate of those who have them. Offer a free 3-month trial of these services to high-risk customers (month-to-month, 0–12 months tenure, no add-ons). Once customers experience the value of these services, conversion to paid subscription is likely and churn risk drops significantly.

### 06 Create a Senior Citizen Support Programme

Senior citizens churn at ~41%, which is notably higher than non-senior customers. This segment may face unique challenges such as difficulty navigating digital billing, lack of tech support, or fixed income sensitivity to pricing. Introduce a dedicated senior support line, simplified billing options, and a senior-specific pricing plan to improve retention in this growing demographic.

## 8. Conclusion

---

This analysis reveals that customer churn at this telecommunications company is not random — it is concentrated in identifiable, addressable segments. Month-to-month contract holders, new customers in their first year, fiber optic subscribers, and electronic check users are all significantly more likely to churn than their counterparts.

The good news is that the data also shows what retention looks like: long-term contracts, add-on service subscriptions, and auto-pay enrolment are all strongly associated with lower churn. These are not accidents — they are levers the business can actively pull.

The six recommendations outlined in this report are directly supported by the data and represent practical, targeted interventions that can be implemented across marketing, product, and customer support teams. Prioritising early-stage retention, long-term contract migration, and service quality for high-value segments would have the greatest measurable impact on reducing the 26.5% churn rate.